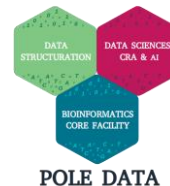




**Storage referential  
based on iRODS,  
for patient data in  
Gustave Roussy  
cancer care center.**

Gérôme JULES-CLEMENT

**GUSTAVE  
ROUSSY**  
CANCER CAMPUS  
GRAND PARIS



**iRODS.**

**UGM 2026**

# Gustave Roussy comprehensive cancer center

- Integrates care, research, and education
- Treats all cancer types, across all ages and disease stages
- Internationally recognized expertise in rare and complex cancers
- Combines medical, scientific, and technological excellence
- Clinical / Translational / Exploratory research context:
  - Large-scale genomic and histology data
    - next-gen sequencing & digital pathology generate terabytes of data
  - Requires HPC infrastructure for advanced analysis
  - Allows retrospective analyses (e.g., digital cohorts)

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## Key data challenges:

- Multiple internal/external data sources
- Diverse technologies and timelines
- Complex clinical context

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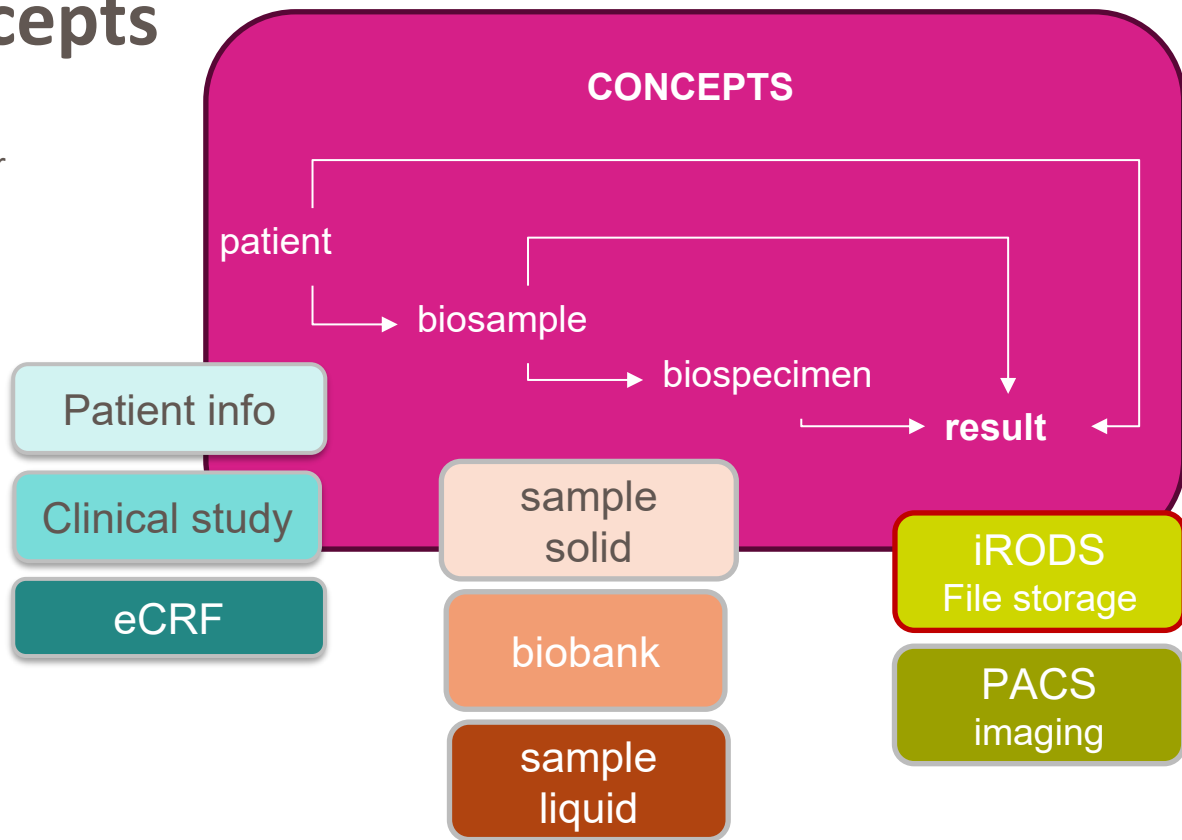


## Institutional solution:

- Centralized storage referential
- Flexible for project-specific analyses
- Built on iRODS + metadata catalog
  - Long-term data preservation
  - Traceability & reuse
  - Secure and controlled access

# Referential concepts

- Dedicated referentials allow modular approach
  - Address specific needs
  - Incremental delivery
  - Allow in house and licenced application
  - Enhance partnership with interoperability
- Project concept corresponds to set of content within each referential





# Referential overview : reflects interoperability

GUSTAVE ROUSSY

UNIVERSITY OF LYON

Biotracking Connect

Last data update : 2026-06-26

Users

JULES-CLEMENT Gerome

MATCH-R

REDCap Project Name

Search

Save

Reset

| NIP       | <div>8 / 16</div> | Birth Date | REDCap                                                                                                              | cBioPortal                               | Glims                                   | Anapath                               | Xenograft | CRB                                                                                                                | iRODS Results                                                                                                                                                      |
|-----------|-------------------|------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------|---------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| c9778d... |                   | 1951       | <a href="#">RCP moléculaire myeloide</a><br><a href="#">SPIDER WEB GR-GYNeco</a><br><a href="#">RCP Moléculaire</a> | <a href="#">GR-HEMADATA</a>              | 2023 (45)<br>2022 (4)<br>2017-2021 (31) | 2023 (1)<br>2022 (1)<br>2017-2021 (5) |           | MATCH-R (10)<br>Comité 060 - Gynécologie (4)                                                                       | Target-seq (1)<br>Exome-seq (3)<br>Single nucleus RNAseq (1)<br>Total RNA-seq (1)<br>Visium FFPE V2 Spatial RNAseq (2)<br>Chromium (1)<br>HE (3)<br>Visium ROI (3) |
| 9e5e09... |                   | 1964       | <a href="#">RCP Moléculaire</a><br><a href="#">UNLOCK</a>                                                           | <a href="#">STING UNLOCK BIOTRACKING</a> | 2023 (89)<br>2022 (5)<br>2019-2021 (10) | 2023 (2)<br>2022 (1)<br>2019-2021 (4) | 2023 (1)  | PANDORE (9)<br>ONCOBIOTICS (9)<br>MATCH-R (21)<br>STING (5)<br>STING-Y1346 (2)<br>Comité 150 - Essais précoces (2) | Target-seq (3)<br>Single nucleus RNAseq (2)<br>Exome-seq (3)<br>mRNA-seq (3)<br>HES (1)                                                                            |

6 - Storage referential based on iRODS, for patient data in Gustave Roussy cancer care center.



# Navigate across referential

**Biotracking Connect**  
Last data update : 2026-06-26

Users  JULES-CLEMENT Gerome 

MATCH-R  REDCap Project Name   Search  Save  Reset

**Sample in DIAMIC of 9e5e097476142785a1cd88bbc414fb23b14249b1** 

| Code  | Sample Date  | CRB Status  | Xenograft  | iRODS Protocol  | cBioPortal  |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| sample_ID51                                                                            | 2023-07-17                                                                                    |                                                                                              |                                                                                             |                                                                                                    |                                                                                                |
| sample_ID52                                                                            | 2023-03-06                                                                                    | Sent,Sent                                                                                    | sample_ PDX1                                                                                | <a href="#">Exome-seq(1), mRNA-seq(1)</a>                                                          | <a href="#">[redacted].I</a> <a href="#">[redacted].R</a>                                      |
| sample_ID53                                                                            | 2022-11-15                                                                                    | Active,Active                                                                                |                                                                                             |                                                                                                    |                                                                                                |
| sample_ID54                                                                            | 2021-04-23                                                                                    | Active                                                                                       |                                                                                             |                                                                                                    |                                                                                                |
| sample_ID55                                                                            | 2021-01-22                                                                                    | Active                                                                                       |                                                                                             | <a href="#">HES(1)</a>                                                                             |                                                                                                |
| sample_ID56                                                                            | 2020-08-03                                                                                    |                                                                                              |                                                                                             |                                                                                                    |                                                                                                |
| sample_ID57                                                                            | 2019-03-28                                                                                    |                                                                                              |                                                                                             |                                                                                                    |                                                                                                |

9e5e09... [UNLOCK](#)  (2) 2019-2021 (10) 2019-2021 (4) STING-Y1346 (2) mRNA-seq (3) Comité 150 - HES (1)

7 - Storage referential based on iRODS, for patient data in Gustave Roussy cancer care center.



# iRODS-Catalog : barcoded storage unit

5 DATASET 4 PATIENT

COLUMNS

EXPORT AS CSV

<

>

Rows per page: 10 1-5 of 5 < >

| DatasetId   | Import date | Biological category | Organisation   | Protocol   | Sample type | Patient                                         | Datafile count             | Depends on id | Sample                       | Project         |
|-------------|-------------|---------------------|----------------|------------|-------------|-------------------------------------------------|----------------------------|---------------|------------------------------|-----------------|
| 2000019<br> | 2020-04-24  | genomics            | IntegraGen     | Exome-seq  | healthy     | patient_ID1<br>patient_aliasA                   | 2 <input type="checkbox"/> | 2000307       | sample_ID10<br>sample_aliasZ | MATCH-R (15)    |
| 2000074<br> | 2020-04-24  | genomics            | IntegraGen     | Exome-seq  | tumor       | patient_ID1<br>patient_aliasA                   | 2 <input type="checkbox"/> | 2000307       | sample_ID90<br>sample_aliasG | MATCH-R (13)    |
| 2006441<br> | 2021-02-16  | transcriptomics     | Novogene       | mRNA-seq   | tumor       | patient_ID2<br>patient_aliasB                   | 2                          |               | sample_ID65<br>sample_aliasJ | MATCH-R (10)    |
| 2040410<br> | 2022-03-11  | genomics            | Gustave Roussy | Target-seq | tumor       | patient_ID3<br>patient_aliasC                   | 1                          | 2040389       | sample_ID22                  | SAFIR_BREAST(0) |
| 2237545<br> | 2026-05-18  | Histology_Slides    | Gustave Roussy | HE         | tumor       | patient_ID4<br>patient_aliasD<br>patient_aliasE | 1                          |               | sample_ID11                  | MEDITWIN (0)    |

8 - Storage referential based on iRODS, for patient data in Gustave Roussy cancer care center.



# iRODS-Catalog : storage unit registered file

Home IRODS - CATALOG RESEARCH FAVORITE 1 JULES-CLEMENT GEROME

5 DATASET 4 PATIENT COLUMNS EXPORT AS CSV Rows per page: 10 1-5 of 5

DatasetId Import Biological Sample Datafile Depends Project

Datafile details (Dataset id: 2000074)

| id     | name                | datafileCategory | biologicalApplication | download                 |
|--------|---------------------|------------------|-----------------------|--------------------------|
| 713670 | -T1-ADN_R2.fastq.gz | raw              | raw genomics          | <input type="checkbox"/> |
| 713671 | -T1-ADN_R1.fastq.gz | raw              | raw genomics          | <input type="checkbox"/> |

|         |            |                  |                |            |       |                                                 |   |         |             |                 |
|---------|------------|------------------|----------------|------------|-------|-------------------------------------------------|---|---------|-------------|-----------------|
| 2040410 | 2022-03-11 | genomics         | Gustave Roussy | Target-seq | tumor | patient_ID3<br>patient_aliasC                   | 1 | 2040389 | sample_ID22 | SAFIR_BREAST(0) |
| 2237545 | 2026-05-18 | Histology_Slides | Gustave Roussy | HE         | tumor | patient_ID4<br>patient_aliasD<br>patient_aliasE | 1 |         | sample_ID11 | MEDITWIN (0)    |

9 - Storage referential based on iRODS, for patient data in Gustave Roussy cancer care center.

# iRODS-Catalog : storage unit registered file

Home IRODS - CATALOG RESEARCH FAVORITE 1 JULES-CLEMENT GEROME

5 DATASET 4 PATIENT COLUMNS EXPORT AS CSV

Project details (Dataset id: 2000019)

| Name                                                                                      | Description      | UserAccess                         |
|-------------------------------------------------------------------------------------------|------------------|------------------------------------|
| MATCH-R  | Project abstract | User list : email gustaveroussy.fr |
| MetaPRISM                                                                                 | Project abstract | User list : email gustaveroussy.fr |
| B21081_LUFR_02                                                                            | Project abstract | User list : email gustaveroussy.fr |

DatasetId Project

2000019 MATCH-R (15)

2000074 MATCH-R (13)

2006441 MATCH-R (10)

2040410 SAFIR\_BREAST(0)

2237545 MEDITWIN (0)

2026-05-18 Histology\_Slides Gustave Roussy HE tumor patient\_aliasD patient\_aliasE 1 sample\_ID11

10 - Storage referential based on iRODS, for patient data in Gustave Roussy cancer care center.



# iRODS-Catalog : data diversity



1 Pb storage

IRODS-CATALOG

RESEARCH

FAVORITE



JULES-CLEMENT GEROME



571 Users



55591 Patients Gustave Roussy



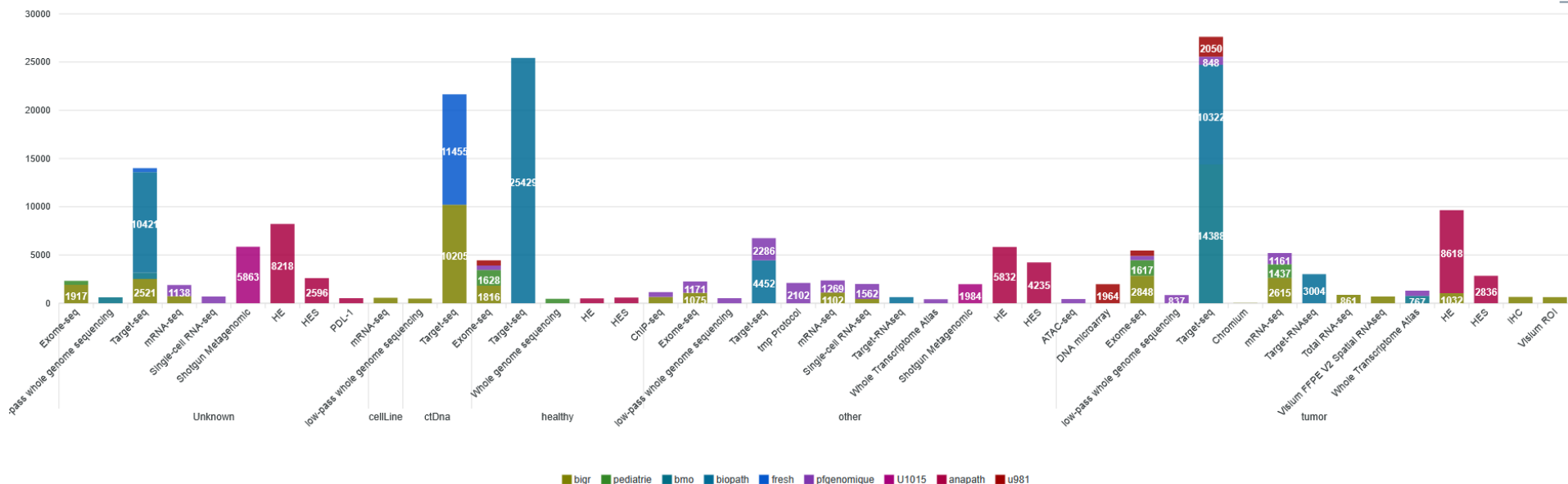
14833 External patients



1967 Projects

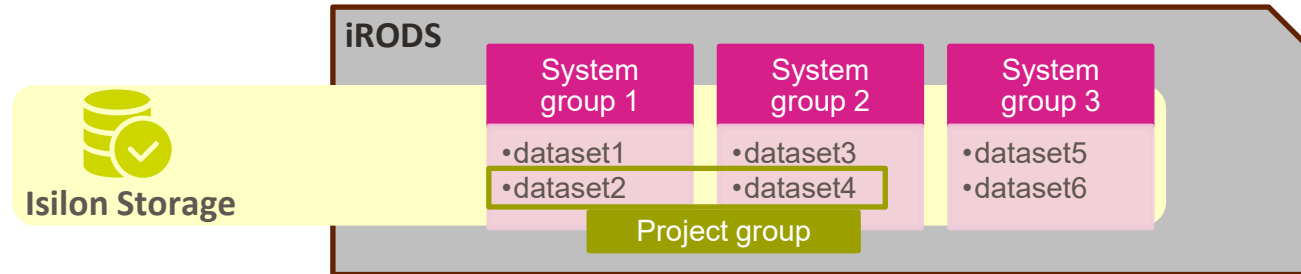


217798 Datasets

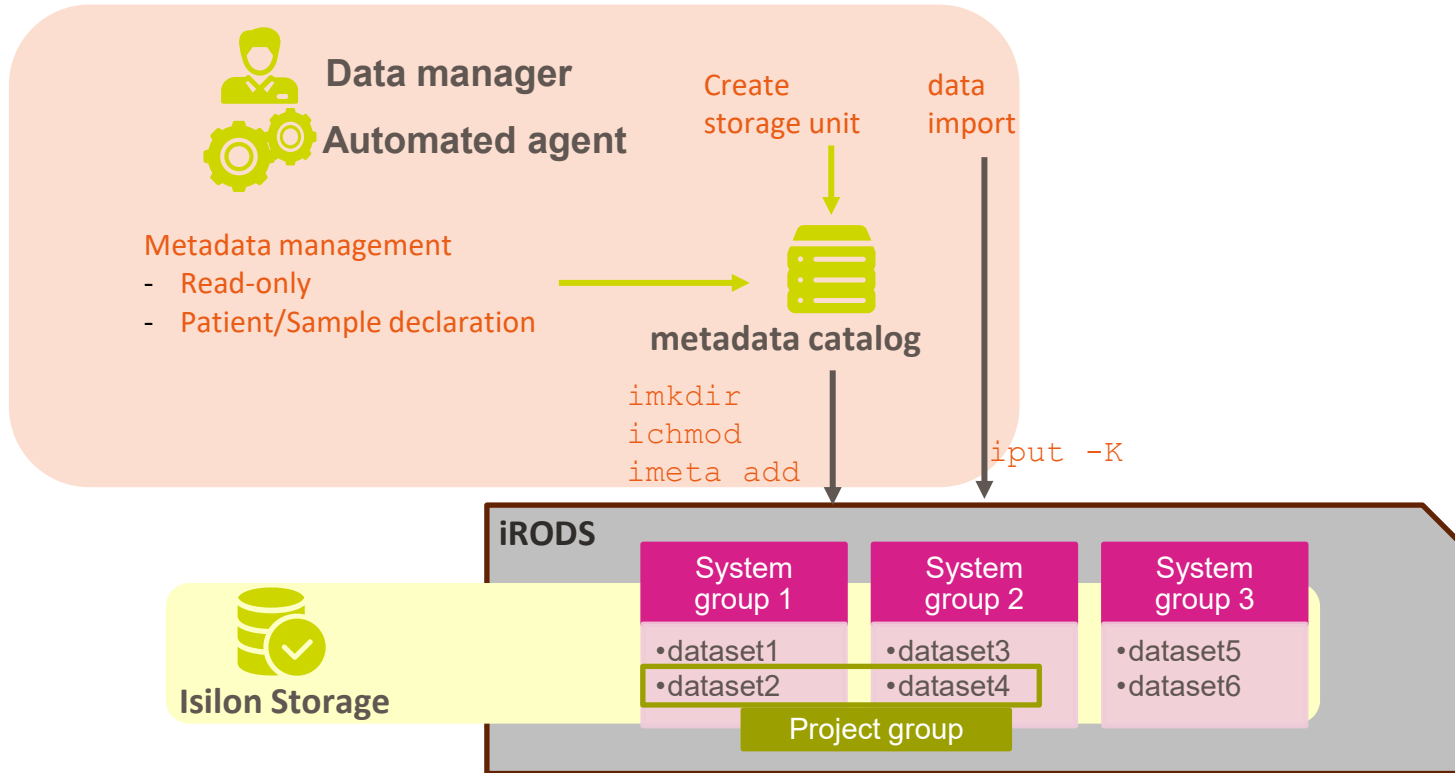


11 - Storage referential based on iRODS, for patient data in Gustave Roussy cancer care center.

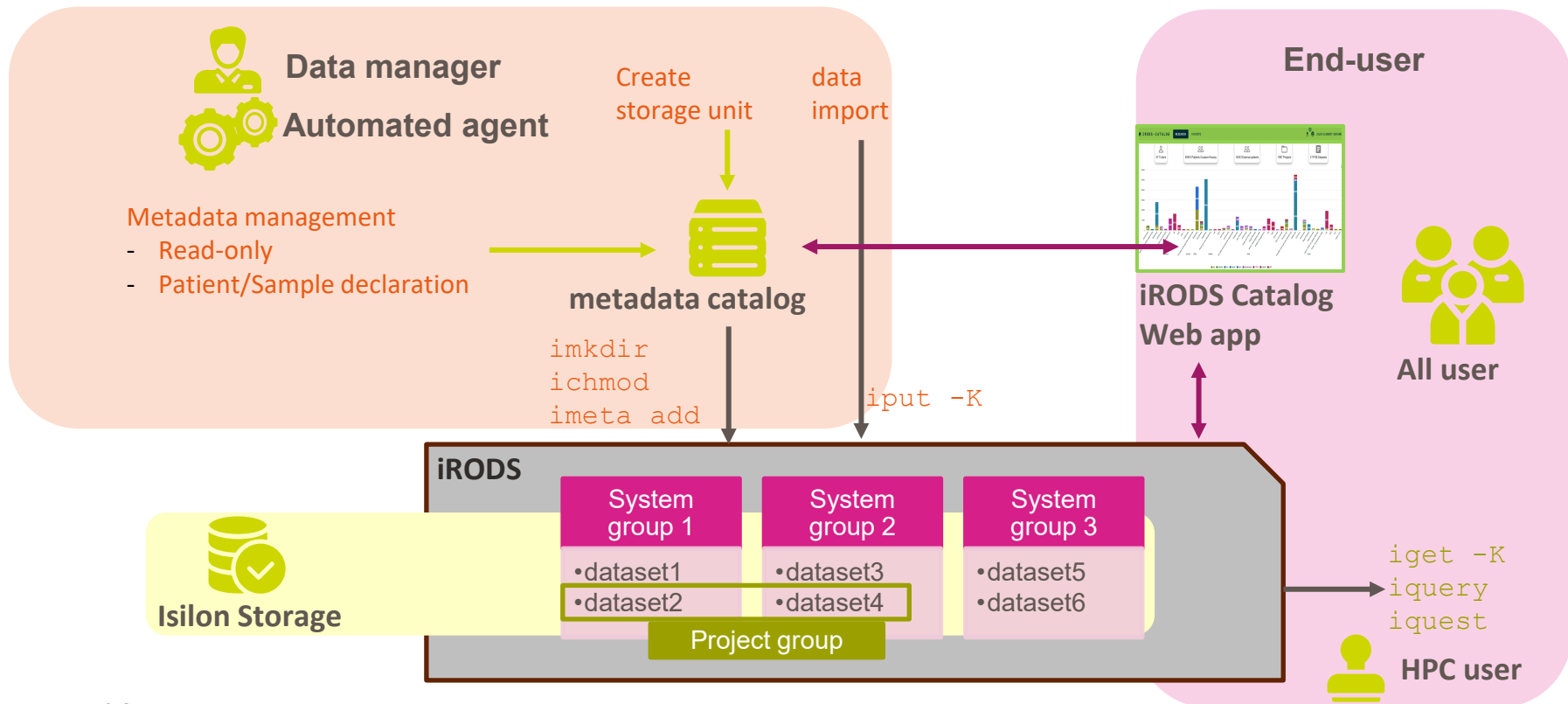
# Storage Referential : iRODS-based system



# Storage Referential : iRODS-based system

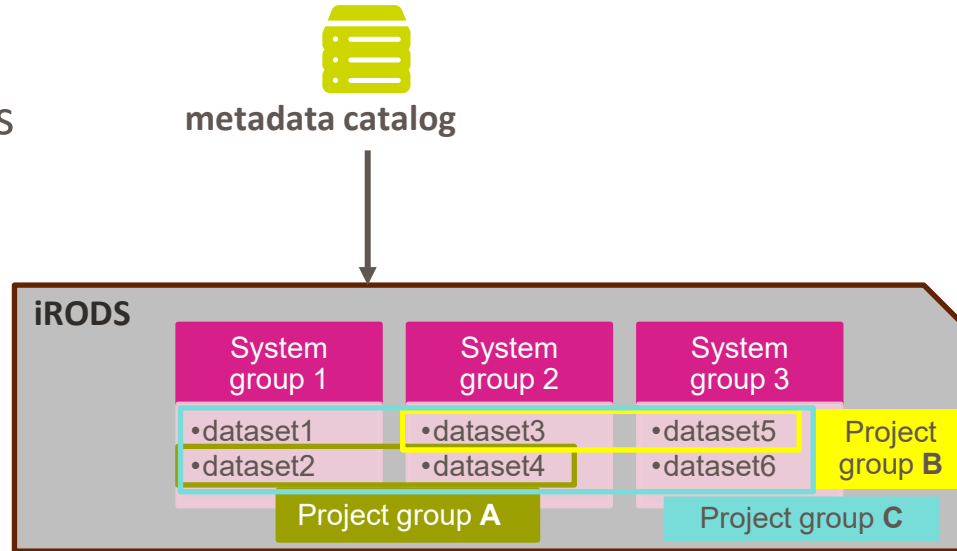


# Storage Referential : iRODS-based system



# Storage Referential : user access configuration

- Create project
  - Name
  - Description
- Add users
- Add storage units



# Storage referential : metadata catalog

## Primary dataset INPUT

```
{
  "organisationName": "Gustave Roussy",
  "datasetIds": [],
  "protocolName": "Exome-seq",
  "tpName": "NovaSeq X Plus",
  "datasetName": "S17765_1",
  "biologicalCategoryName": "genomics",
  "projectName": "GR_PROJECT"
}
```

## Analysis dataset INPUT :

```
{
  "organisationName": "Gustave Roussy",
  "datasetIds": [<primary datasetId>],
  "analysisTypeName": "https://gitlab.com##<commit-hash>",
  "datasetName": "analysis1",
  "biologicalCategoryName": "genomics",
  "projectName": "GR_RETROSPECITE_PROJECT"
}
```



# Storage referential : metadata catalog

## Primary dataset INPUT

```
{
  "organisationName": "Gustave Roussy",
  "datasetIds": [],
  "protocolName": "Exome-seq",
  "tpName": "NovaSeq X Plus",
  "datasetName": "S17765_1",
  "biologicalCategoryName": "genomics",
  "projectName": "GR_PROJECT"
}
```

**OUTPUT** is a storage unit identifier : datasetId  
/odin/kdi/dataset/<datasetId>

## Analysis dataset INPUT :

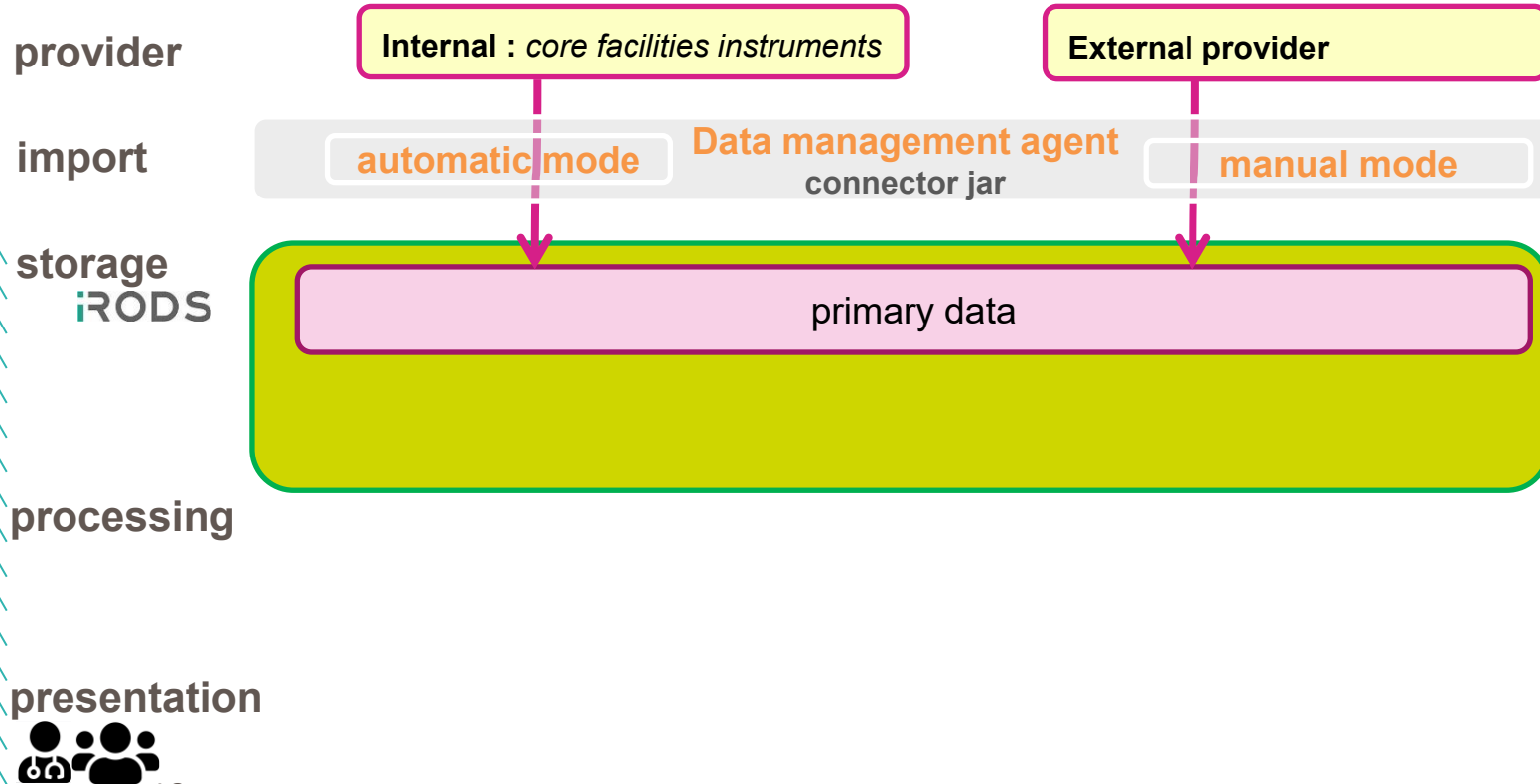
```
{
  "organisationName": "Gustave Roussy",
  "datasetIds": [<primary datasetId>],
  "analysisTypeName": "https://gitlab.com##<commit-hash>",
  "datasetName": "analysis1",
  "biologicalCategoryName": "genomics",
  "projectName": "GR_RETROSPECITE_PROJECT"
}
```

# Inter-operability with patient/sample referential

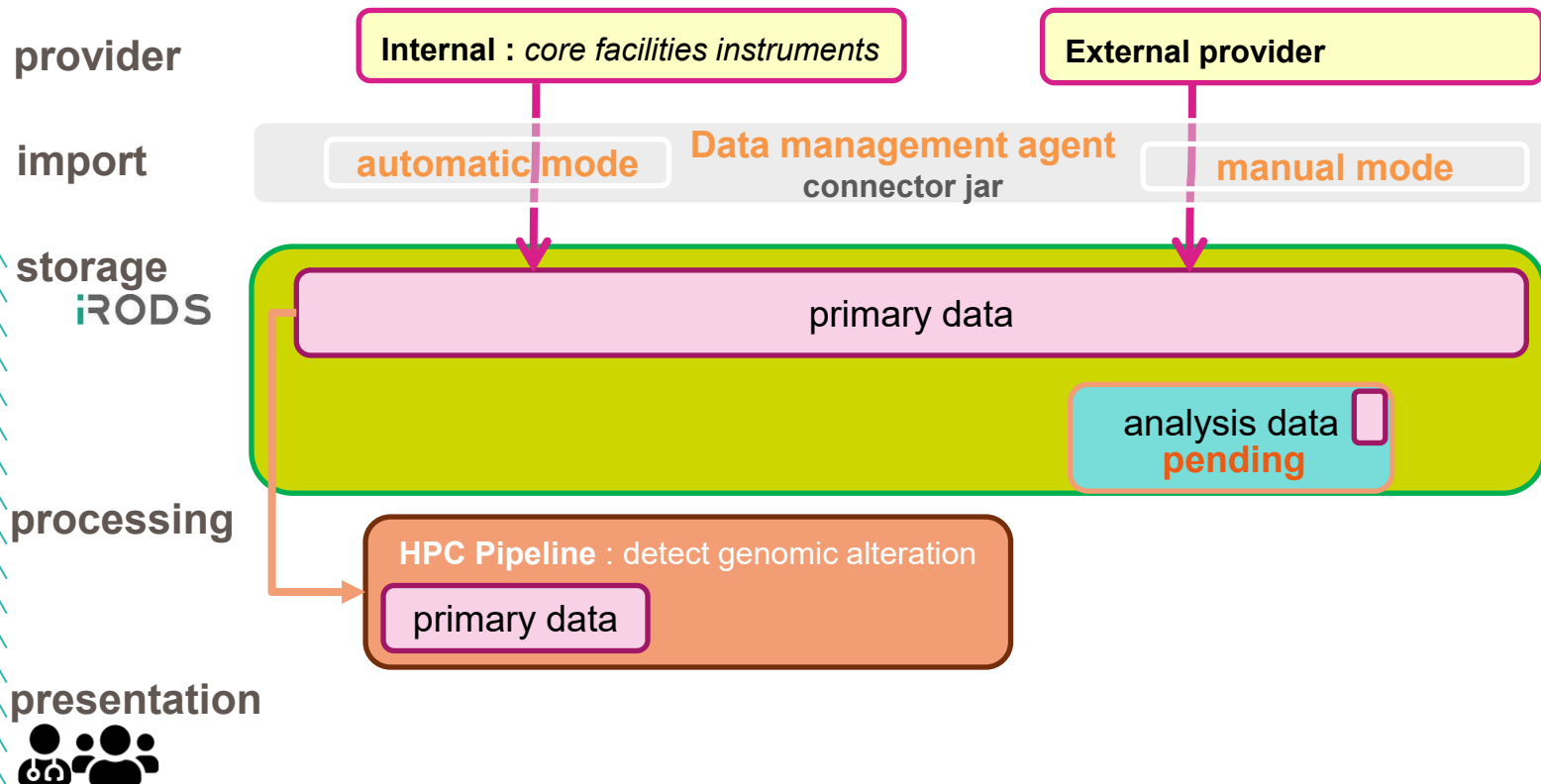
Sample declaration template:

```
{
  "sampleType": "tumor",
  "speciesName": "Homo sapiens",
  "sampleNameType": "Sample number",
  "sampleName": "S12345",
  "sampleAliases": [],
  "comment": ""
  "patientGender": "female",
  "patientNameType": "NOIGR",
  "patientName": "PatientID1",
  "patientAliases": [
    {
      "typeName": "Patient inclusion number",
      "value": "P12345"
    }
  ],
}
```

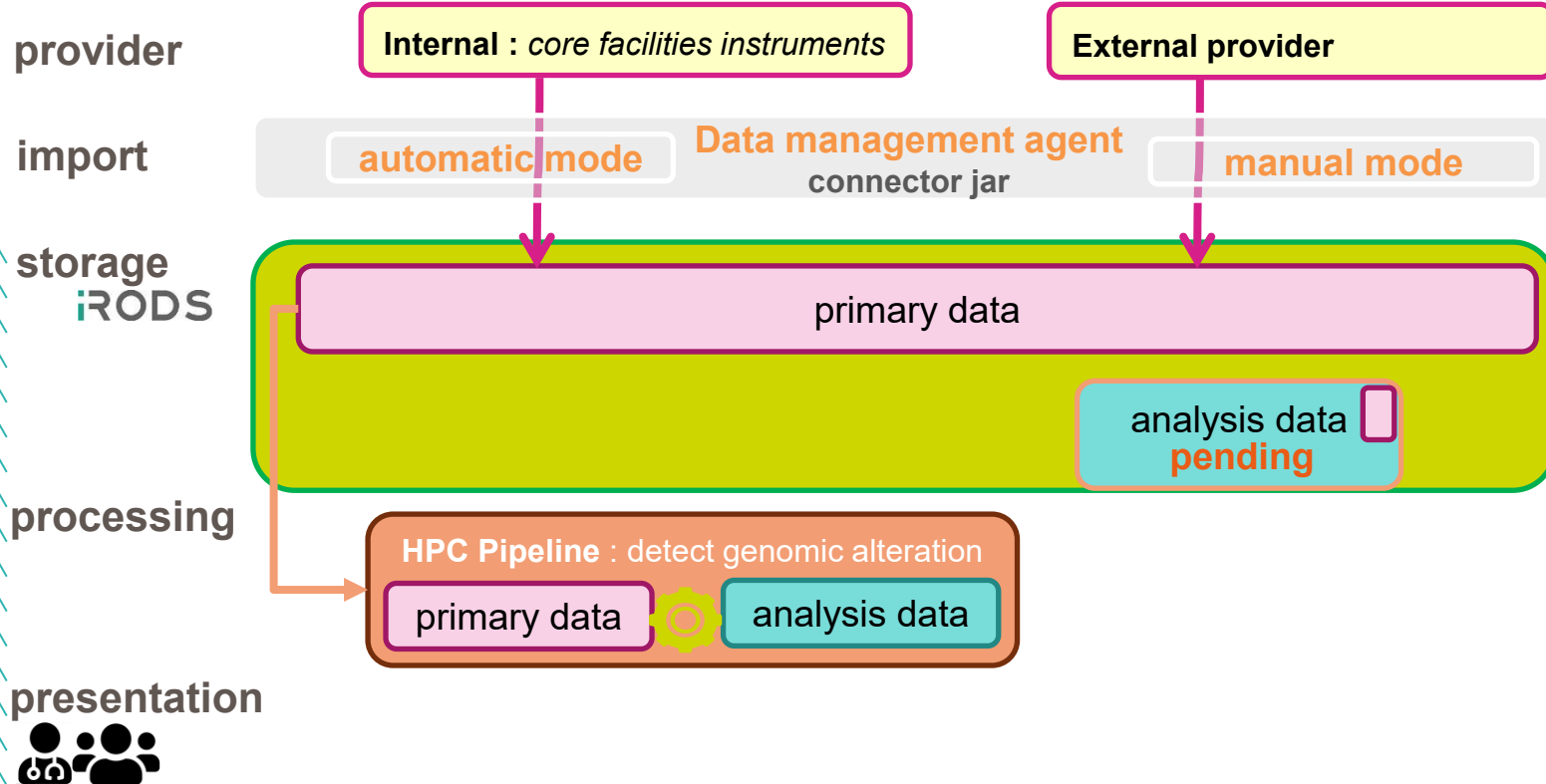
# Data life cycle: example of genomics data



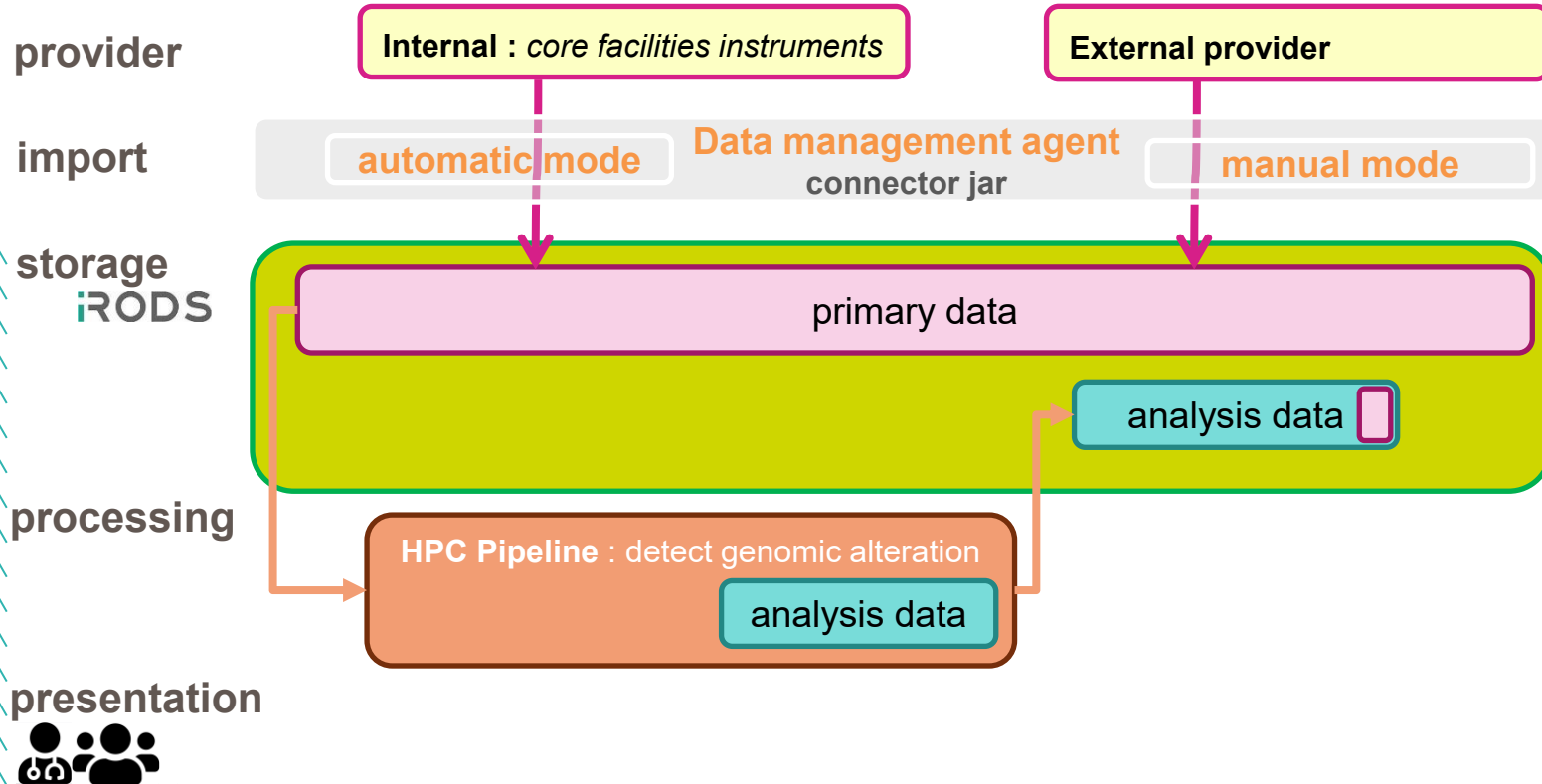
# Data life cycle: example of genomics data



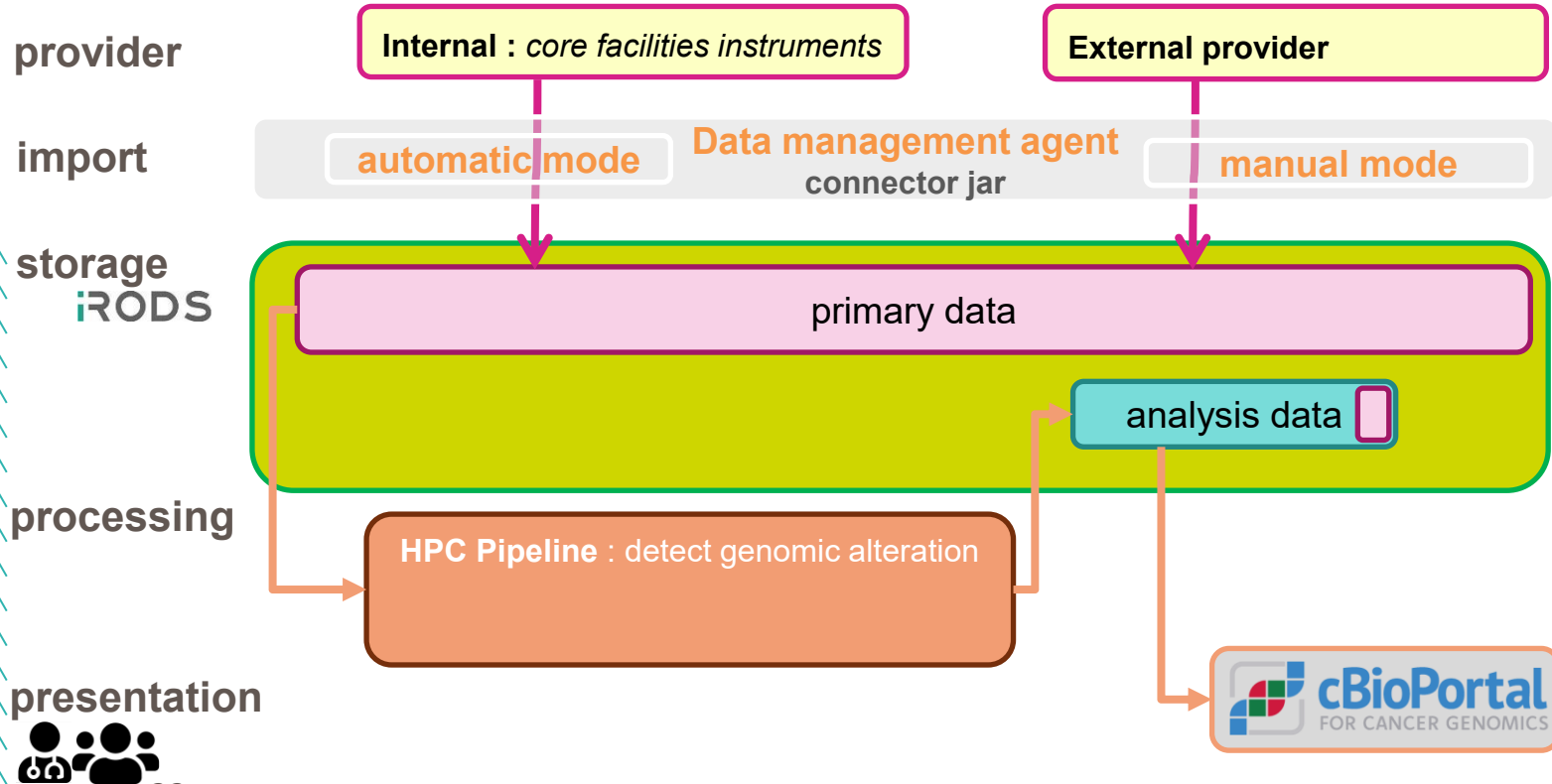
# Data life cycle: example of genomics data



# Data life cycle: example of genomics data



# Data life cycle: example of genomics data



# Conclusion



Biomedical data growth requires more than simple file storage

- 6+ years of production use proves reliability
  - Powered by iRODS 4.2.7 in 2020 then iRODS 5.0.2 in 2026
- Centralizes heterogeneous data under unified governance
- Reduces workload via automation of data management
- Promotes FAIR principles (Findable, Accessible, Interoperable, Reusable)



Gustave Roussy developed a scalable, long-term storage referential

- Designed for omics and digital pathology data management
- Supports both routine clinical workflows and large-scale research
- Enables retrospective cohort building and advanced studies
- Improves collaboration between clinicians and researchers





**Thank you !**

**Questions?**